Bangladesh University of Engineering and Technology

DEPARTMENT OF BIOMEDICAL ENGINEERING

Course Outline

Course No: BME 211
Course Title: Bioelectricity
Credit hours: 3 **Contact hours:** 3
Level/Term: 2/2
Academic Session: 2023-2024

Course Teacher(s):

Name	Office/Room	Email	Class routine
Samiul Based Shuvo	ECE-1121	sbshuvo@bme.buet.ac.bd +880 1521258181	Sunday: 09:00 AM-09:50 AM Monday: 09:00 AM-09:50 AM Tuesday: 12:00 PM-12:50 PM
Tasnia Binte Mamun	ECE-1123	tasnia@bme.buet.ac.bd +880 1677-900345	

Course Contents:

Introduction to Bioelectricity and Excitable Cells; Bioelectric potentials and currents: ionic composition of excitable cells, Nernst-Planck equation, membrane structure, Nernst potential, parallel-conductance model; Membrane channels: channel structure, biophysical methods for measuring channel properties, macroscopic channel kinetics, channel statistics, introduction to the Hodgkin-Huxley membrane model; Action potentials: observing action potentials, nonlinear membrane behavior, origin of action potential, resting and peak voltages, voltage and space clamp, Hodgkin-Huxley equations, simulation of membrane action potential, action potential characteristics, active transport, calcium channels and "other" membrane models; Impulse propagation: core-conductor model, cable equations, local circuit currents during propagation, mathematics of propagating action potentials, propagation velocity constraint for uniform fiber, propagation in myelinated nerve fibers; Electrical stimulation of excitable

tissue: linear (subthreshold) response of a single spherical cells, linear (subthreshold) response of a cylindrical fiber; Cardiac electrophysiology: electrical nature of intercellular communication, source models, ECG measurement and analysis; The Neuromuscular junction: structure of the neuromuscular junction, evidence for the quantal nature of transmitter release, Poisson statistics for transmitter release, the effect of Ca^{2+} and Mg^{2+} on transmitter release, post-junctional response to transmitter; Skeletal muscle: muscle structure, muscle contraction, structure of the myofibril, sliding filament theory, Excitation-contraction, EMG measurement and analysis; Neural electrophysiology: structure of the nervous system, sensory transducers and neurons, neural synapses, excitation and inhibition, neural coding and computation, EEG measurement and analysis, brain-computer interfaces; Functional electrical stimulation: electrodes and electrode-tissue behavior, nerve excitation; Recent advancements in Bioelectricity.

Course Objectives:

- The course will enable students to model the membrane of an excitable cell and cellular phenomena
- The course will help the students to learn fundamental knowledge of bio-signal generation and propagation
- The course will enable the students to analyze design characteristics and parameters of electrical stimulation
- The course will demonstrate state-of-the-art research applications correlating different diseases and conditions relevant to bioelectricity

Course Outcomes:

1. Understand the fundamental basis of bioelectric phenomena and bio-signal generation
2. Investigate characteristics of membrane/channels/ cellular phenomena by constructing appropriate mathematical models
3. Analyze the generation and propagation of surface electrical potentials
4. Construct bioelectrical system that meets specified requirements

Assessment:

Class participation/Attendance:	10%
Homework Assignment and Quizzes:	20%

Term Final Exam: 70%

Textbooks and References:

1. R. Plonsey and R. C. Barr, “Bioelectricity: A Quantitative Approach,” 3rd Edition, Springer, 2007.
2. D. Johnston and S. M.-S. Wu, “Foundations of cellular neurophysiology,” MIT Press, 1994.
3. R. M. Gulrajani, “Bioelectricity and Biomagnetism,” John Wiley &. Sons, Inc. New York, 1998.
4. C. Koch, “Biophysics of Computation: Information Processing in Single Neurons,” Oxford University Press, 1998.
5. J. Feher, “Quantitative Human Physiology: An Introduction”, 2nd Edition, Elsevier, 2017.
6. D. Silverthorn, “Human Physiology: An Integrated Approach”, 7th Edition, Pearson, 2016.

Weekly Schedule:

Week	Topics
1	Introduction to bioelectricity and excitable cells. Basic biophysical concepts: sources and fields. Current applications, future trends, and challenges of bioelectricity.Skeletal muscle: Muscle structure; muscle contraction; structure of the myofibril; Sliding filament theory, Excitation-contraction.
2	The neuromuscular junction: Structure of the neuromuscular junction; Evidence for the quantal nature of transmitter release, Poisson statistics for transmitter release, the effect of Ca^{2+} and Mg^{2+} on transmitter release, Parallel-Conductance Model of Postsynaptic Membrane; post-junctional response to transmitter.
3	EMG measurement and analysis; EMG-controlled movement of prosthetic hand/foot. Cardiac electrophysiology: Electrical nature of intercellular communication; gap-junctional structure, functional continuity and resistance, Cardiac cellular models, idealized temporal and spatial cardiac action potential waveforms.
4	ECG measurement and analysis; Neural electrophysiology: Structure of the nervous system; sensory vs motor neurons; Neural coding and computation
5	Neural electrophysiology: Origin of EEG; EEG measurement and analysis; Brain-computer interface (BCI)
Mid-term break	
6	Functional electrical stimulation: Electrodes and electrode-tissue behavior; electrode material and types; Electrode operating characteristics; nerve

	excitation; Different types of stimulation based on application and corresponding waveforms.
7	Analysis of electrode performance and criteria for satisfactory performance. Recent advancements in Bioelectricity: Pain reducing and rehabilitation; Emotion recognition; cognitive study; sleep disorder and seizure analysis. Synthetic speech generation from EEG; gesture/eye movement/EEG-controlled movement of prosthetic hand/foot.
8	Bioelectric potentials and currents: Ionic composition of excitable cells; ionic flux; diffusion and Fick's Law; Ionic mobility, conductance and conductivity; Einstein equation; Nernst-Planck equation; Nernst potential.
9	Bioelectric potentials and currents: Membrane structure and properties; Single-ion permeability: concentration cell, Transmembrane potential and currents, Nernst Equilibrium; Modeling membrane characteristics: Parallel-conductance model; chloride contribution; effect of sudden reduction in external chloride concentration.
10	Membrane channels: Channel structure and types; selective permeability; active and passive transport through channels; Channel gating: Boltzmann distribution; Biophysical methods for measuring channel properties: Patch-clamp configurations, workflow and considerations for its fair use ; Macroscopic channel kinetics; channel statistics; Introduction to the Hodgkin-Huxley membrane model.
11	Action potentials: Origin of action potential; resting and peak voltages; action potential characteristics; Voltage and space clamp techniques for action potential measurement; Recording action potential: noteworthy attributes; Hodgkin-Huxley equations; Nonlinear membrane behavior; mathematical model and equations for Na, K; Simulation of membrane action potential.
12	Impulse propagation: Core-conductor model; cable equations; Local circuit currents during propagation, propagation in myelinated nerve fibers; Mathematics and simulation of propagating action potentials; Propagation velocity constraint for uniform fiber.
13	Electrical stimulation of excitable tissue: Crucial simulation parameters; strength-duration curve; Linear(subthreshold) response of a single spherical cell; Linear(subthreshold) response of a cylindrical fiber; fiber equations.

Prepared by:	
Name: Signature: Date:	Tasnia Binte Mamun

Name: Signature: Date:	Samiul Based Shuvo